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batch-46

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| <b>SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE</b>                             |                   | <b>DEPARTMENT OF COMPUTER SCIENCE ENGINEERING</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                  |
| <b>Program Name:</b> B. Tech                                                              |                   | <b>Assignment Type:</b> Lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Academic Year:</b> 2025-2026  |
| <b>Course Coordinator Name</b>                                                            |                   | Dr. Rishabh Mittal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                  |
| <b>Instructor(s) Name</b>                                                                 |                   | <div> <div>Mr. S Naresh Kumar</div> <div>Ms. B. Swathi</div> <div>Dr. Sasanko Shekhar Gantayat</div> <div>Mr. Md Sallauddin</div> <div>Dr. Mathivanan</div> <div>Mr. Y Srikanth</div> <div>Ms. N Shilpa</div> <div>Dr. Rishabh Mittal (Coordinator)</div> <div>Dr. R. Prashant Kumar</div> <div>Mr. Ankushavali MD</div> <div>Mr. B Viswanath</div> <div>Ms. Sujitha Reddy</div> <div>Ms. A. Anitha</div> <div>Ms. M.Madhuri</div> <div>Ms. Katherashala Swetha</div> <div>Ms. Velpula sumalatha</div> <div>Mr. Bingi Raju</div> </div> |                                  |
| <b>CourseCode</b>                                                                         | 23CS002PC304      | <b>Course Title</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | AI Assisted Coding               |
| <b>Year/Sem</b>                                                                           | III/II            | <b>Regulation</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | R23                              |
| <b>Date and Day of Assignment</b>                                                         | Week8 – Wednesday | <b>Time(s)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 23CSBTB01 To 23CSBTB52           |
| <b>Duration</b>                                                                           | 2 Hours           | <b>Applicable to Batches</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | All batches                      |
| <b>Assignment Number:</b> 16.3(Present assignment number)/24(Total number of assignments) |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                  |
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| <b>Q.No.</b>                                                                              | <b>Question</b>   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Expected Time to complete</b> |



|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                    |
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| 1 | <p>Lab 16 – Database Design and Queries: Schema Design and SQL Generation</p> <p>Lab Objectives:</p> <ul style="list-style-type: none"><li>• Understand schema design principles for relational databases.</li><li>• Use AI-assisted tools to generate SQL schema definitions.</li><li>• Write and refine SQL queries (CRUD + aggregate operations).</li><li>• Explore how AI can optimize queries and ensure normalization.</li></ul> <hr/> <p>Task 1 – Hospital Management Database Task:<br/>Create schema and queries for a Hospital Management System.</p> <p>Instructions:</p> <ul style="list-style-type: none"><li>• Tables: Doctors, Patients, Appointments.</li><li>• Use AI to define constraints (unique IDs, valid dates).</li><li>• Generate queries:</li><li>• List all appointments for a specific doctor.</li><li>• Retrieve patient history by patient ID.</li><li>• Count total patients treated by each doctor.</li></ul> <p>Expected Output:</p> <ul style="list-style-type: none"><li>• Normalized schema and SQL queries with joins.</li></ul> <p>CODE:</p> <p>DROP TABLE IF EXISTS Appointments;<br/>DROP TABLE IF EXISTS Patients; DROP<br/>TABLE IF EXISTS Doctors;</p> <p>CREATE TABLE Doctors (<br/>  doctor_id INT PRIMARY KEY,<br/>  name VARCHAR(100),<br/>  specialization VARCHAR(100)<br/>);</p> <p>CREATE TABLE Patients (<br/>  patient_id INT PRIMARY KEY,<br/>  name VARCHAR(100),<br/>  age INT,<br/>  gender VARCHAR(10)<br/>);</p> <p>CREATE TABLE Appointments (<br/>  appointment_id INT PRIMARY KEY,<br/>  doctor_id INT,   patient_id INT,<br/>  appointment_date DATE,<br/>  FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id),<br/>  FOREIGN KEY (patient_id) REFERENCES Patients(patient_id)<br/>);</p> | Week8<br>Wednesday |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|

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|--|---------------------------------------------------------------------------|--|
|  | <div>INSERT INTO Doctors VALUES<br/>(1, 'Dr. Smith', 'Cardiology'),</div> |  |
|--|---------------------------------------------------------------------------|--|

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre>(2, 'Dr. John', 'Neurology');  INSERT INTO Patients VALUES (101, 'Alice', 25, 'Female'), (102, 'Bob', 30, 'Male');  INSERT INTO Appointments VALUES (1, 1, 101, '2025-08-01'), (2, 1, 102, '2025-08-02'), (3, 2, 101, '2025-08-03');  SELECT * FROM Doctors; SELECT * FROM Patients; SELECT * FROM Appointments;  SELECT  a.appointment_id,  a.appointment_date,  p.name  AS patient_name FROM Appointments a JOIN Patients p ON a.patient_id = p.patient_id WHERE a.doctor_id = 1;  SELECT  a.appointment_id,  a.appointment_date,  d.name  AS</pre> |  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

```
doctor_name
FROM Appointments a
JOIN Doctors d ON a.doctor_id = d.doctor_id
WHERE a.patient_id = 101;

SELECT d.name, COUNT(DISTINCT a.patient_id) AS total_patients
FROM Doctors d
LEFT JOIN Appointments a ON d.doctor_id = a.doctor_id
GROUP BY d.name;
```

Task 2 – Real-Time Application: Online Shopping Database Scenario:  
Design a database for an E-commerce platform.

Requirements:

- Tables: Users, Products, Orders, OrderDetails.
- Generate queries to:
- Retrieve all orders by a user.
- Find the most purchased product.
- Calculate total revenue in a given month.
- AI should also suggest normalization improvements and query optimization.

Expected Output:

- Complete schema with relationships + SQL queries for analytics.

CODE:

```
DROP TABLE IF EXISTS OrderDetails;
```

```

DROP TABLE IF EXISTS Orders;
DROP TABLE IF EXISTS Products;
DROP TABLE IF EXISTS Users;

CREATE TABLE Users (
  user_id INT PRIMARY KEY,
  name VARCHAR(100),
  email VARCHAR(100) UNIQUE
);

CREATE TABLE Products (
  product_id INT PRIMARY KEY,
  name VARCHAR(100),
  price DECIMAL(10,2)
);

CREATE TABLE Orders (
  order_id INT PRIMARY KEY,
  user_id INT, order_date DATE,
  FOREIGN KEY (user_id) REFERENCES Users(user_id)
);

CREATE TABLE OrderDetails (
  order_detail_id INT PRIMARY KEY,
  order_id INT, product_id INT,
  quantity INT,
  FOREIGN KEY (order_id) REFERENCES Orders(order_id),
  FOREIGN KEY (product_id) REFERENCES Products(product_id)
);

INSERT INTO Users VALUES
(1, 'Alice', 'alice@email.com'),
(2, 'Bob', 'bob@email.com');

INSERT INTO Products VALUES
(1, 'Laptop', 50000),
(2, 'Phone', 20000);

INSERT INTO Orders VALUES
(1, 1, '2025-08-01'),
(2, 1, '2025-08-10'),
(3, 2, '2025-08-15');

INSERT INTO OrderDetails VALUES
(1, 1, 1, 1),
(2, 2, 2, 2),
(3, 3, 2, 1);

```

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|





```

-- 1. Orders by user
SELECT * FROM Orders WHERE user_id = 1;

-- 2. Most purchased product
SELECT p.name, SUM(od.quantity) AS total_sold
FROM OrderDetails od
JOIN Products p ON od.product_id = p.product_id
GROUP BY p.name
ORDER BY total_sold DESC
LIMIT 1;

-- 3. Total revenue (August 2025)
SELECT SUM(p.price * od.quantity) AS total_revenue
FROM Orders o
JOIN OrderDetails od ON o.order_id = od.order_id
JOIN Products p ON od.product_id = p.product_id
WHERE MONTH(o.order_date) = 8 AND YEAR(o.order_date) =
2025;

```

### Task 3 – Library Database Task:

Design schema for a Library Management System.

Instructions:

- Tables: Books, Members, Loans.
- Use AI to suggest indexing strategy for faster queries.
- Generate queries:
- Retrieve all books currently issued.
- Find overdue books (loan date > 30 days).
- Count number of books loaned by each member.

Expected Output:

- Schema + SQL queries demonstrating joins and conditions.

CODE:

```

DROP TABLE IF EXISTS Loans;
DROP TABLE IF EXISTS Members;
DROP TABLE IF EXISTS Books;

```

```

CREATE TABLE Books (
  book_id INT PRIMARY KEY,
  title VARCHAR(100),
  author VARCHAR(100)
);

```

|  |                                                                                                   |  |
|--|---------------------------------------------------------------------------------------------------|--|
|  | <pre>CREATE TABLE Members (<br/>  member_id INT PRIMARY KEY,<br/>  name VARCHAR(100)<br/>);</pre> |  |
|--|---------------------------------------------------------------------------------------------------|--|

```
CREATE TABLE Loans (
```



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre>loan_id INT PRIMARY KEY, book_id INT, member_id INT, loan_date DATE, return_date DATE, FOREIGN KEY (book_id) REFERENCES Books(book_id), FOREIGN KEY (member_id) REFERENCES Members(member_id) );  INSERT INTO Books VALUES (1, 'DBMS', 'Author A'), (2, 'AI Basics', 'Author B');  INSERT INTO Members VALUES (1, 'John'), (2, 'Alice');  INSERT INTO Loans VALUES (1, 1, 1, '2025-07-01', NULL), (2, 2, 2, '2025-06-01', NULL);  -- 1. Books currently issued SELECT b.title, m.name FROM Loans l JOIN Books b ON l.book_id = b.book_id JOIN Members m ON l.member_id = m.member_id WHERE l.return_date IS NULL;  -- 2. Overdue books (&gt;30 days) SELECT b.title, m.name FROM Loans l JOIN Books b ON l.book_id = b.book_id JOIN Members m ON l.member_id = m.member_id WHERE DATEDIFF(CURDATE(), l.loan_date) &gt; 30 AND l.return_date IS NULL;  -- 3. Books per member SELECT m.name, COUNT(l.book_id) AS total_books FROM Members m LEFT JOIN Loans l ON m.member_id = l.member_id GROUP BY m.name;</pre> |  |
|  | <p>Task 4: Optimize Queries Instructions:</p> <ul style="list-style-type: none"><li>• Use AI tools to analyze query efficiency.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |

- |  |                                                                                                                                                                                |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <ul style="list-style-type: none"><li>• Optimize inefficient queries using joins, indexes, or reducing subqueries.</li><li>• Test optimized queries for correctness.</li></ul> |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

Starter Code Example: -

- Original query



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre>SELECT * FROM Books WHERE author_id IN (SELECT author_id FROM Authors WHERE country='UK');</pre> <p>-- Optimized query</p> <pre>SELECT b.*<br/>FROM Books b<br/>JOIN Authors a ON b.author_id = a.author_id<br/>WHERE a.country = 'UK';</pre> <p>Expected Output:</p> <ul style="list-style-type: none"><li>• Both queries return the same result: all books written by UK authors.</li><li>• Optimized query performs faster for larger datasets.</li></ul> <p>CODE:</p> <pre>SELECT * FROM Books<br/>WHERE author_id IN (<br/>    SELECT author_id FROM Authors WHERE country='UK'<br/>);</pre> <pre>SELECT b.*<br/>FROM Books b<br/>JOIN Authors a ON b.author_id = a.author_id<br/>WHERE a.country = 'UK';</pre>                                                                                                                                                   |  |
|  | <p>Task 5: University Course Registration Scenario:</p> <p>SR University needs a course registration system for students enrolling in different courses under faculty supervision.</p> <ul style="list-style-type: none"><li>• Use Copilot to design schema tables: Students, Courses, Faculty, and Registrations.</li><li>• Define relationships:</li><li>• Each student can register for multiple courses.</li><li>• Each course is taught by a faculty member.</li><li>• Write SQL queries to:</li><li>• List all students enrolled in a specific course.</li><li>• Find faculty members teaching more than 2 courses.</li><li>• Retrieve courses with the highest number of registrations.</li></ul> <p>CODE:</p> <pre>DROP TABLE IF EXISTS Registrations;<br/>DROP TABLE IF EXISTS Courses;<br/>DROP TABLE IF EXISTS Faculty;<br/>DROP TABLE IF EXISTS Students;</pre> |  |



|  |                                                                                                     |  |
|--|-----------------------------------------------------------------------------------------------------|--|
|  | <pre>CREATE TABLE Students (<br/>  student_id INT PRIMARY KEY,<br/>  name VARCHAR(100)<br/>);</pre> |  |
|--|-----------------------------------------------------------------------------------------------------|--|

```
CREATE TABLE Faculty (  
  faculty_id INT PRIMARY KEY,  
  name VARCHAR(100)  
);  
  
CREATE TABLE Courses (  
  course_id INT PRIMARY KEY,  
  course_name VARCHAR(100),  
  faculty_id INT,  
  FOREIGN KEY (faculty_id) REFERENCES  
  Faculty(faculty_id)  
);  
  
CREATE TABLE Registrations (  
  registration_id INT PRIMARY KEY,  
  student_id INT,  
  course_id INT,  
  FOREIGN KEY (student_id) REFERENCES
```

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
|  | <pre> Students(student_id),     FOREIGN KEY (course_id) Courses(course_id) );  INSERT INTO Students VALUES (1, 'Alice'), (2, 'Bob'), (3, 'Charlie');  INSERT INTO Faculty VALUES (1, 'Prof. A'), (2, 'Prof. B');  INSERT INTO Courses VALUES (101, 'DBMS', 1), (102, 'AI', 1), (103, 'Networks', 2);  INSERT INTO Registrations VALUES (1, 1, 101), (2, 2, 101), (3, 3, 102), (4, 1, 102), (5, 2, 103);  -- 1. Students in a course SELECT s.name FROM Registrations r JOIN Students s ON r.student_id = s.student_id WHERE r.course_id = 101; </pre> | REFERENCES |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
|  | <pre>-- 2. Faculty teaching &gt;2 courses SELECT f.name, COUNT(c.course_id) AS total_courses FROM Faculty f JOIN Courses c ON f.faculty_id = c.faculty_id GROUP BY f.name HAVING COUNT(c.course_id) &gt; 2;  -- 3. Course with highest registrations SELECT      c.course_name, COUNT(r.student_id) total_students FROM Courses c JOIN Registrations r ON c.course_id = r.course_id GROUP BY c.course_name ORDER BY total_students DESC LIMIT 1;</pre> | AS |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|